PRACTICAL 6 : Interface 16X2 LCD with Raspberry Pi using 12C module to display text

REQUIRMENT:

Raspberry Pi 400

LCD 1602 with I2C module

4 Female-to-Female jumper wires

🔌 **Wiring (I2C LCD)**

LCD Pin	Connect to Raspberry Pi GPIO
GND	Pin 6 (GND)
VCC	Pin 4 (5V)
SDA	Pin 3 (GPIO2 - SDA)
SCL	Pin 5 (GPIO3 - SCL)

Diagram:

LCD Pi 400 GPIO
---- ----------
GND --> GND (Pin 6)
VCC --> 5V (Pin 4)
SDA --> SDA (GPIO 2 / Pin 3)
SCL --> SCL (GPIO 3 / Pin 5)

🖥️ **Step 1: Enable I2C on Raspberry Pi**

sudo raspi-config

Go to Interface Options → I2C → Enable

Reboot your Pi

📦 **Step 2: Install Required Packages**

sudo apt update

sudo apt install i2c-tools python3-smbus python3-pip

pip3 install RPLCD

Check if the LCD is detected:

i2cdetect -y 1

You'll see something like

0 1 2 3 4 5 6 7 8 9 A B C D E F

00: -- -- -- -- -- -- -- --

10: -- -- -- -- -- -- -- -- -- -- -- --

20: -- -- -- -- -- -- -- -- -- 27 -- --

Note down the address (usually 0x27 or 0x3F).

🧪 **Step 3: Test LCD with Python**

CODE:

```
from RPLCD.i2c import CharLCD
from time import sleep
lcd = CharLCD('PCF8574', 0x27) # Change 0x27 if needed
lcd.write_string("Hello from Pi 400!")
sleep(3)
lcd.clear()
```

Save this as lcd_test.py and run:

python3 lcd_test.py

✅ **Your LCD should now display the message!**

